A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 38

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 110 x 0.67 x 1.35 x (0.50 x 0.90) x 1.0 **1** = 44.8 t ha^{-1} yr^{-1} (accept 44.7-44.8) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (t ha^{-1} yr^{-1}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 110 x 0.67 x 1.35 x (0.35 x 0.60) x 0.25 **1** = 5.2 t ha^{-1} yr^{-1} **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (t ha^{-1} yr^{-1}) is already printed on the answer line. • **contour ploughing** → ploughing along the slope rather than up and down it **Do not credit** • A bare wrong answer with no working.	2
01.3	**1 mark each** 1 mark for each of the 3 points below. **1 Beans / up-and-down ploughing:** severe class (over 30) and about 45 times the formation rate - clearly unsustainable, soil depth will fall **1** Wheat / mulch / strip cropping: moderate class, reduced by about 88% but still 5 times the formation rate so not fully sustainable **1** Further measures (long-term crops, terracing, windbreaks, more organic matter) or grazing land (C = 0.02) needed to bring loss below 1 t ha^{-1} yr^{-1} **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs • **terracing** → cutting the slope into level steps **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Factors are averages / arbitrary values for typical conditions – rainfall erosivity varies between years / storms **1** Does not include gully erosion, landslides, wind erosion or local variation in soil – based on data from particular regions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.5	1 mark for the point below. **1** C (crop / tillage) or P (erosion control practice) - R, K and LS are fixed by climate, soil and topography **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 167-169; AQA spec 3.2.5.2.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (210 - 12) / 210 x 100 **1** = 94.3% (accept 94%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. • **recycling** → reprocessing a material to make something new **Do not credit** • A bare wrong answer with no working. • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	2
02.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Conserves mineral reserves / extends their lifetime **1** Reduces the environmental impacts of mining (land take, habitat loss, spoil, dust, noise, turbid water) **1** Reduces mineral processing pollution (eg red mud from bauxite refining) **1** Reduces waste disposal impacts / landfill volume **1** Less CO2 / climate change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3

Do not write outside the box

Question	Marking guidance	Marks
02.3	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Identification of materials:** slow, labour-intensive to determine composition where it matters (eg alloys, plastic types) **1 Separation of mixed materials:** alloys cannot easily be separated so recycled aluminium cannot be used where pure metal is needed – mixed wastes reduce quality **1** Reduction in quality / contamination (eg paper mixed with food waste) **1 Transport / collection:** small dispersed quantities cannot use bulk transport so costs and energy per tonne are higher (may exceed savings) **1** Lack of consumer cooperation in separating wastes – waste losses to litter and general waste – labour costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	4
02.4	1 mark for the point below. **1** Designing products so that the materials can be re-used at the end of their lives - easy separation of components and identification of materials - extending the lifetime of mineral use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 146-147; AQA spec 3.2.3.7.	
03.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (68 000 - 7 000) / 68 000 x 100 **1** = 89.7% (accept 90%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Inflow from the rivers was reduced (sometimes to zero) so less water entered while evaporation continued **1** Evaporation removes pure water leaving the dissolved salts behind, so they became concentrated in a smaller volume **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**1 mark each** 1 mark for each of the 4 points below. **1** Salinity rose above the range of tolerance of the freshwater fish species / osmotic stress – pollution by pesticides, fertilisers and industrial waste concentrated in the lake **1** Exposed lake bed of salt and pesticide-contaminated dust blown by wind onto farmland and settlements (health problems, soil salinisation) **1** Loss of wetland habitats and wildlife – local climate more extreme (loss of the moderating effect of the water body) **1** Loss of livelihoods – conflict between the countries sharing the rivers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **range of tolerance** → the range of conditions in which a species can survive • **salinisation** → salt building up in the soil / soil becoming salty **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Upstream abstraction for irrigation reduces the flow available downstream **1** Upstream HEP dams hold water in summer for winter generation, reducing summer irrigation water downstream – growing populations increase demand **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abstraction** → water taken out of a river, lake or aquifer • **HEP** → electricity generated from falling water **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117-118; AQA spec 3.2.2.1-3.2.2.2.	
04.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Steam washing (hot water / steam jets kill organisms in the sediment)	1

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Chemicals cause the oil molecules to join together / solidify **1** Producing more solid material that can be collected / lifted more easily than liquid oil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Air contains about 210 cm^3 of oxygen per litre whereas water at 20 °C holds a maximum of about 6 cm^3 per litre (less than 3% of the amount in air) **1** So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen – many aquatic organisms are near the lower end of their range of tolerance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water • **range of tolerance** → the range of conditions in which a species can survive	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1 Mutagen:** an agent that changes the chemical structure / base sequence of DNA (causes mutations), eg ionising radiation, UV, PCBs, cadmium, asbestos **1 Carcinogen:** a mutagen that causes cancer / uncontrolled cell division producing a tumour **1 Teratogen:** an agent that causes birth abnormalities by preventing normal gene expression (inhibiting proteins / enzymes) without changing DNA structure, eg mercury, 2,4,5-T **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **ionising radiation** → radiation with enough energy to remove electrons and form ions **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Teratogens do not change the DNA structure, so the abnormality is not in the genes and cannot be passed on **1** Mutagens change DNA – if the mutation is in a gamete / egg / sperm (gonadic effect) it can be inherited **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1 Somatic:** mutation in a general body cell, affecting the individual only (eg cancer / tumour) **1 Gonadic:** mutation in cells of the ovaries or testes / gametes / embryo, which may cause abnormalities in offspring **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.4	**1 mark each** 1 mark for each of the 3 points below. **1** Two or more pollutants whose effects interact to produce a different (usually more serious) effect **1** The pollutants themselves do not react to form a new pollutant (that would be a secondary pollutant) **1 Example:** ozone damages leaf cuticles so sulfur dioxide can cause more damage to the exposed cells – smoke and sulfur dioxide in smogs – neonicotinoids with fungicides **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **synergism** → two pollutants together doing more damage than the sum of each alone	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 239-240; AQA spec 3.4.1.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** Each regional population has its own unique gene pool / DNA profile **1** DNA from seized timber or ivory is compared with the database of regional populations **1** This identifies where the material came from, showing whether it was legally harvested and where poaching occurs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	3

Do not write outside the box

Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Radio / GPS transmitters use radio waves - work in air and fresh water, giving positions (GPS stores or transmits location) **1** Radio waves do not travel through seawater so acoustic transmitters (sound) are used for marine animals, giving real-time positions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
06.3	**Marks shared** Up to 3 marks in total - share them between the 2 points below according to how completely each one is made. **up to 2** Drones / ROVs, AUVs (eg Autosub), balloons, aircraft, satellites, animals (tags) **up to 2 Drones:** cheap, low disturbance, high-resolution images of areas hard to reach (cliffs, canopy, ice), repeated surveys – infrared cameras detect animals at night **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Cameras record automatically (visible light by day, infrared at night) without researchers being present **1** So animals behave naturally and nests are not deserted – monitoring continues 24 hours **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-406; AQA spec 3.7.2.3.	
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - rivers are accessible over a large area with few natural contaminants	1

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 2 points below. **1 Potable / public supply:** no pathogens, low toxins, good appearance / taste / smell (public health) **1 Industrial boiler / cooling water:** soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange – spray irrigation: low turbidity to avoid blocked pipes and low heavy metals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) **1 Disadvantage:** high cost / energy for pumping – ecological impacts on the donor river – transfer of pollutants or invasive species – political conflict **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	
08.1	**1 mark each** 1 mark for each of the 3 points below. **1** A turbofan at the front forces bypass air around the inner cowling / core jet **1** The bypass air smooths the flow of the turbulent hot exhaust gases, reducing the turbulence that creates noise **1** The higher the ratio of bypass air to jet exhaust the quieter the engine (Trent 1000 quietest, Olympus noisiest) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
08.2	**Any 2 of 6** 1 mark each for any 2 of the 6 points below, in any order, maximum 2. **1** Chevron (serrated) nozzles mix bypass air with surrounding air more smoothly **1** Acoustic liners in the engine cowlings absorb noise **1** Hush kits fitted to older low-bypass engines **1** Fairings on undercarriage and wing-flap hinges reduce turbulence – welded rather than riveted panels **1** Lighter composite aircraft need less thrust **1** Blended wing with engines on top so the body shields the ground **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
08.3	**1 mark each** 1 mark for each of the 4 points below. **1** Constant (3°) descent avoids periods of high thrust in a stepped descent and keeps aircraft higher for longer **1** Night flights (especially take-offs, which are noisier) are restricted / banned when people are sleeping and background noise is low **1** Noisier aircraft pay higher charges / use up more quota points and fines for exceeding limits, so airlines are encouraged to buy quieter aircraft **1** More time to slow down after landing so wheel brakes can be used instead of noisy reverse thrusters **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quota** → a legal limit on the mass or number that may be caught **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4
08.4	1 mark for the point below. **1** Engines run at higher power / maximum thrust and the steep climb angle needs full power (although the aircraft climbs quickly out of earshot) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 288-292; AQA spec 3.4.3.2.13.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 2 points below. **1** Natural processes may concentrate the material (eg adsorption onto sediments, bioaccumulation and food-chain concentration) **1** It may be carried to locations where it is hazardous, such as fishing grounds or beaches used by people **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorption** → nutrient ions sticking to the surfaces of clay and humus • **bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
09.2	**Any 4 of 8** 1 mark each for any 4 of the 8 points below, in any order, maximum 4. **1** Physical state of the effluent (solution / solid) **1** Ocean currents / river flow **1** Effect of pH and oxygen on solubility **1** Bioaccumulation and food-chain concentration **1** Food sources and consumption by local people (eg shellfish, seaweed) **1** Half-lives of the isotopes **1** Meteorological conditions for any gaseous release **1** Nature of the environment - geology, vegetation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain	4
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Monitoring focuses on the sites most at risk of contamination **1** No need to sample sites that were never likely to be polluted – sampling checks the accuracy of the predictions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accuracy** → close to the true value	2

Do not write outside the box

Question	Marking guidance	Marks
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** It assesses the risk to the people most exposed because of their lifestyle before any health effects occur **1** If their risk is acceptably low everyone else has a lower risk – if unacceptable, emissions are reduced to prevent future exposure **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Critical Group** → the people most at risk / most heavily exposed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
10.1	**1 mark each** 1 mark for each of the 3 points below. **1** Together they form a dispersed layer in the stratosphere (the ozone layer / ozonosphere) **1** They absorb ultraviolet light, which drives a dynamic equilibrium of reactions that form and destroy ozone **1** The UV energy is converted to chemical / heat energy so most UV that passed the upper atmosphere never reaches the surface **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3
10.2	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Provides carbon, oxygen, hydrogen and nitrogen (as N_2, O_2, CO_2, H_2O) for biological molecules **1** Delays the escape of infrared energy, keeping the surface warm **1** Distributes heat and water vapour by winds / ocean currents **1** Provides atmospheric pressure so liquid water can exist **1** Deflects damaging radiation of the solar wind in the upper atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
10.3	1 mark for the point below. **1** Energy is absorbed at different altitudes (eg UV absorbed in the stratosphere, surface heating warms the troposphere) so temperature gradients change with altitude, producing layers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **stratified** → sampling each area or group in proportion to its size	1
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Troposphere:** enhanced greenhouse effect - anthropogenic CO_2 (fossil fuels, deforestation), methane (rice, livestock, landfill), N_2O, CFCs, tropospheric ozone – more infrared absorbed and re-emitted – aerosols and smoke increase albedo / scatter light – land-use change alters surface albedo and evapotranspiration • **Stratosphere:** CFCs / halons / HCFCs release chlorine and bromine which catalyse ozone destruction, shifting the O / O_2 / O_3 equilibrium – less UV absorbed in the stratosphere, more reaches the surface – stratospheric cooling • **Consequences:** changes to climate, ocean currents (thermohaline), hydrological cycle, sea level, species distributions – increased UV damage to organisms • **Difficulties of prediction:** linked processes where one change alters others – feedback (positive and negative) and tipping points – time lags (ocean heat capacity, long residence times of CFCs and CO_2) – natural variability (solar output, volcanoes, Milankovitch cycles) – incomplete understanding of atmospheric processes – limited historical data so proxies needed – models with different assumptions • **Evaluation:** some effects well understood (greenhouse physics, ozone chemistry proven, Montreal Protocol success) but scale and timing of regional impacts uncertain – precautionary principle • Justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

Do not write outside the box

Question	Marking guidance	Marks
	• **enhanced greenhouse effect** → extra warming caused by added greenhouse gases • **anthropogenic** → human-caused / caused by human activity • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem	
	Model answer - Level 3 (8 marks)	
	In the troposphere human activity enhances the greenhouse effect: fossil fuels and deforestation add CO_2, rice, livestock and landfill add methane, fertilisers add nitrous oxide, and CFCs and tropospheric ozone are new absorbers. More outgoing infrared is absorbed and re-emitted, warming the surface, while smoke and aerosols scatter sunlight and land-use change alters albedo and evapotranspiration. In the stratosphere, CFCs, halons and HCFCs release chlorine and bromine that catalyse ozone destruction, shifting the O / O_2 / O_3 equilibrium so that less UV is absorbed there and more reaches the surface, cooling the stratosphere as well.	
	The consequences run through every system the atmosphere's energy drives: climate, thermohaline circulation, the hydrological cycle, sea level and species distributions, plus UV damage to organisms. Predicting them is hard because the processes are linked - one change alters others - and because feedback can amplify (albedo loss, methane release) or damp (cloud cover, photosynthesis) the effect, sometimes past tipping points. Time lags from ocean heat capacity and the long residence times of CO_2 and CFCs, natural variability from solar output, volcanoes and orbital cycles, sparse historical data replaced by proxies, and models built on different assumptions all add uncertainty.	
	Some things are nevertheless well established: the physics of infrared absorption, the chemistry of ozone loss, and the success of the Montreal Protocol. What is uncertain is the scale and timing of regional impacts, which is exactly why the precautionary principle, not certainty, should guide action.	
	Weaker answer - Level 1 (3 marks)	
	Humans release greenhouse gases like carbon dioxide from burning fossil fuels into the troposphere, which traps more heat and causes global warming. CFCs go into the stratosphere and destroy the ozone layer, so more UV reaches the Earth. It is hard to predict what will happen because the atmosphere is complicated and scientists do not know everything about it. There could be feedback effects that make it worse.	
	What makes the difference	
	• Level 1 gives one cause per layer; Level 3 lists several gases and processes and explains the mechanism (absorption and re-emission; catalytic destruction shifting an equilibrium).	
	• 'Difficulties of prediction' is answered in Level 3 with specific reasons (linked processes, feedback, time lags, natural variability, proxy data, models); Level 1 says 'complicated'.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3 evaluates - separating what is certain from what is not - and links this to the precautionary principle.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89, 108-113; AQA spec 3.2.1.1, 3.2.1.3.	
11.1	**Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Composition and structure:** N_2, O_2, Ar, CO_2, ozone – troposphere and stratosphere – pressure and liquid water – gases for biological molecules and industry • **Energy processes:** UV absorbed by O / O_2 / O_3 dynamic equilibrium – visible light absorbed at the surface – infrared absorbed by greenhouse gases (33 °C warming) – heat distribution by winds and ocean currents – transport of water vapour – energy budget and albedo • **Global climate change:** anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species – difficulties of monitoring and prediction – control strategies and agreements • **Ozone depletion:** CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • **Other threats:** aerosols, smoke, land-use change – interconnection of processes	25

Do not write outside the box

Question	Marking guidance	Marks
	• **Evaluation:** which functions are most at risk – success of ozone controls vs slow progress on greenhouse gases – justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	
11.2	**Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Attributes:** accuracy, precision, representative samples, reliability, anomalies, validity – hypothesis and null hypothesis – preliminary studies • **Sampling design:** random vs systematic location and avoiding bias – timing (diurnal, seasonal, long-term, weather) – sample size and number and the running mean – standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • **Statistics:** variability, standard deviation and error bars, significance and p-values, correlation vs causation – choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • **Application to Paper 1 topics:** monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	

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